

Efficient Testing for the strongly connectivity of digraphs

Hsuan-Chang Chen*

Yuh-Dauh Lyuu[†]

Yen-Wu Ti[‡]

April 3, 2007

Abstract

We construct an efficient randomized algorithm that, given a digraph, tests if it is a strongly connected graph. And the query complexity of our algorithm is polylogarithmic in the size of the digraph.

1 Introduction

A classic computational problem is to verify if an object has a predetermined property (e.g., the commutability of a group). Unfortunately, sometimes no fast algorithms exist that give exact answers. In those situations, an approximate answer within a reasonable complexity can be an attractive alternative. A property-testing algorithm offers one such alternative. It determines

whether a problem instance has a certain property or is far from any instances having such a property. Its complexity is measured by the number of queries.

This paper studies property testing for the strongly connectivity of digraphs. An ϵ -tester for a digraph property Π is a randomized algorithm that is given a size parameter n , a distance parameter ϵ and the ability to make queries as to whether a directed edge of the input digraph G with n vertices exists. The total number of queries is called the query complexity of the tester. Let $\{g_i\}$ be the set of digraphs with n vertices that satisfy Π . The algorithm needs to distinguish with probability at least $2/3$ between the case of G having Π and the case of G differing from any g_i in at least $\epsilon \binom{n}{2}$ edges [1]. In the latter case, G is said to be ϵ -far from property Π . The probability $2/3$ can be replaced by any constant smaller than 1 because the algorithm can be repeated if necessary. And a graph property is testable if the property has a tester, and the total number of queries is $o(n^2)$.

A graph property is testable if the property has a tester, and the total number of queries is $o(n^2)$. The testing of graph properties was pioneered by Goldreich, Goldwasser and Ron [12]. They showed that all graph properties describ-

*Dept. Computer Science and Information Engineering, National Taiwan University, Taipei, Taiwan. Email: r93117@csie.ntu.edu.tw.

[†]Dept. Computer Science Information and Engineering, National Taiwan University, No.1, Sec. 4, Roosevelt Road, Taipei, Taiwan 106. Email: lyuu@csie.ntu.edu.tw. The author was supported in part by NSC grant 95-2213-E-002-044.

[‡]Dept. Computer Science Information and Engineering, National Taiwan University, Taipei, Taiwan. Email: d91010@csie.ntu.edu.tw. The author was supported in part by NSC grant 95-2213-E-002-044.

able by the existence of a partition of a certain type are testable. For a fixed digraph H with at least one edge, let P_H denote the property of the input digraph being H -free. In other words, the digraph G has P_H if and only if it contains no subgraphs isomorphic to H . Alon and Shapira proved that the property of H -freedom is testable with the total number of queries bounded only by a function of ϵ , independent of n [1]. This result has been improved later [2]. Alon, Fischer, Krivelevich and Szegedy showed that every first-order undirected graph property without a quantifier alternation of type " $\forall\exists$ " has ϵ -testers whose query complexity is independent of the size of the input digraph [4]. The testing of other graph properties and combinatorial properties has been intensively studied [6, 7, 11, 13, 15].

2 The Main Result

We say that a digraph is strongly connected if there is a directed path from every vertex u in the digraph to every other vertex v . In this section we develop a tester on the strong connectivity of digraphs.

To begin with, testing strong connectivity is trivial when the distance parameter is bigger than $2/n$ for the following reason: We can always make a digraph connected by adding at most $n - 1$ edges. Hence every digraph with n vertices is $(2/n)$ -close to being strongly connected because $(2/n)\binom{n}{2} = n - 1$. On the other hand, ϵ should be bigger than $1/\binom{n}{2}$ for, otherwise, $\epsilon\binom{n}{2} < 1$. To make sure that the problem is not trivial, we assume $1/\binom{n}{2} < \epsilon < 2/n$ from now on.

A *strongly connected component* of a digraph $G = (V, E)$ is a maximal subgraph $H = (V', E')$

such that there is a directed path from each vertex $p \in V'$ to every other vertex $q \in V'$. Denote the set of strongly connected components of G by $C = \{C_1, C_2, \dots, C_m\}$. For vertices $u \in C_i$ and $v \in C_j$, $i \neq j$, such that $e = (u, v) \in E$, we call e an *outgoing edge* of C_i and an *incoming edge* of C_j . Call a strongly connected component a *source* if it has only outgoing edges, a *sink* if it has only incoming edges, an *isolation* if it has neither outgoing nor incoming edges, and a *transferrer* if it has both outgoing and incoming edges.

Lemma 2.1. *If a digraph G with n vertices is ϵ -far from being strongly connected, then the total number of sources, sinks, and isolations in G exceeds ϵn^2 .*

Proof: Assume the claim of the lemma is wrong and proceed to obtain a contradiction. Let set $\mathcal{T}$ contain all transferrers and set $\mathcal{R}$ consisting of the remaining strongly components. We divide the problem into two cases.

Case 1. $\mathcal{T}$ is empty.

A strongly connected component in G is either a source, a sink, or an isolation, and hence a member of $\mathcal{R}$. Pick a vertex from each strongly connected component of G . As G has $|\mathcal{R}|$ strongly connected components, $|\mathcal{R}|$ vertices are chosen. Turn these $|\mathcal{R}|$ vertices into a directed cycle by adding at most $|\mathcal{R}|$ directed edges to G . Now, for any ordered pair of vertices (u, v) in V , there is a directed path from u to v ; hence G is strongly connected. Recall that an ϵ -far digraph differs from the class of strongly connected digraphs by at least $\epsilon n^2 + 1$ edges. But this is a contradiction because $|\mathcal{R}| \leq \epsilon n^2$.

Case 2. $\mathcal{T}$ is not empty.

A chain of strongly connected components $(C_1, C_2, \dots, C_n)$ consists of strongly connected components such that between any two adjacent strongly connected components C_i and C_{i+1} , there is a directed edge from a vertex of C_i to a vertex of C_{i+1} .

For any $T \in \mathcal{T}$, there is a longest chain of strongly connected components S in $\mathcal{T}$ containing T (i.e., all members of S are transferrers, and T is one of them). Let S^h be the head of the chain S . S^h is a strongly connected component with both incoming and outgoing edges because $S^h \in \mathcal{T}$. There is no directed edge from other strongly connected components in $\mathcal{T}$ to S^h for, otherwise, S would not be the longest chain containing T . Therefore all the incoming edges of S^h are outgoing edges of some sources. By the same token, all the outgoing edges of the tail S^t of S are incoming edges of some sinks. Consequently, for any $T \in \mathcal{T}$, we can always find a vertex p from a source and a vertex q from a sink such that there is a directed path from p to q passing through a vertex of T . As a result, if all the sources and sinks are turned into one single strongly connected component, the transferrers will become part of the same component, too.

Therefore, if we can turn all members of $\mathcal{R}$ into one single strongly connected component $\mathcal{R}'$, then for all $T_i, T_j \in \mathcal{T}$ and all vertices $x \in T_i$ and $y \in T_j$, there exists a directed path from x through some vertices in $\mathcal{R}'$ to y . So we can ignore the transferrers and concentrate on how to make all members of $\mathcal{R}$ become one single strongly connected subgraph. We have thus reduced this case to case 1, where $\mathcal{T}$ is empty. The

number of directed edges we need to add to G is thus at most $|\mathcal{R}|$, too. If $|\mathcal{R}|$ is no larger than ϵn^2 , we get a contradiction. ■

Lemma 2.2. *A digraph ϵ -far from being strongly connected must have at least ϵn connected components each containing fewer than $\lceil \frac{1}{n-2}(\frac{1}{\epsilon} - 1) \rceil$ vertices.*

Proof: Assume the claim is wrong. Lemma 3.3 says that this digraph has at least $\epsilon n^2 + 1$ strongly connected components. Since every strongly connected component has at least one vertex,

$$\begin{aligned} n &\geq (\epsilon n^2 + 1 - \epsilon n) \cdot \left\lceil \frac{1}{n-2} \left(\frac{1}{\epsilon} - 1 \right) \right\rceil + \epsilon n \cdot 1 \\ &> (\epsilon n^2 - \epsilon n) \frac{1}{n-2} \left(\frac{1}{\epsilon} - 1 \right) + \epsilon n \\ &= \epsilon n(n-1) \frac{1}{n-2} \left(\frac{1}{\epsilon} - 1 \right) + \epsilon n \\ &= \frac{n-1}{n-2} (n - \epsilon n) + \epsilon n \\ &> n - \epsilon n + \epsilon n \\ &= n, \end{aligned}$$

a contradiction. ■

The algorithm that tests whether a digraph is strongly connected appears in Fig. 1. Recall that test strong connectivity is trivial when the distance parameter is bigger than $2/n$. To make sure that the problem is not trivial, we assume $1/\binom{n}{2} < \epsilon < 2/n$. The following theorem analyzes the algorithm.

Theorem 2.3. (1) *Our algorithm has one-sided error.* (2) *If G is ϵ -far from being strongly connected, then our algorithm will reject it with probability at least $2/3$.* (3) *The query complexity is $O\left(\frac{1}{\epsilon \log(1-\epsilon)}\right)$.*

CONN($\epsilon, n, G = (V, E)$) :

```

1: Let  $S = \emptyset$ ,  $m = \lceil \log_{1-\epsilon} 1/3 \rceil$ ,  $x = \lceil \frac{1}{n-2}(\frac{1}{\epsilon} - 1) \rceil$ 
2: while  $|S| < m$  do
3:   Pick an arbitrary vertex  $u$  from  $V$  and add it to  $S$ 
4:   Perform BFS on  $G$  starting from  $u$  and always using a visited vertex's incoming edges
5:   Stop when we reach  $x$  vertices (including  $u$ ) or run out of new vertices
6:   if we run out of new vertices then
7:     REJECT
8:   else
9:     Perform BFS starting from  $u$  and always using a visited vertex's outgoing edges
10:    Stop when we reach  $x$  vertices (including  $u$ ) or run out of new vertices
11:    if we run out of new vertices then
12:      REJECT
13:    end if
14:  end if
15: end while
16: ACCEPT

```

Figure 1: An ϵ -tester for the connectivity of digraphs.

Proof: Since $1/\binom{n}{2} < \epsilon < 2/n$, $x \leq n$. For a strongly connected digraph G , given an arbitrary vertex u of V , a BFS always reaches at least x vertices by starting from u . In other words, our algorithm never rejects a YES instance; hence it has one-sided error.

For a NO instance G , according to Lemma 2.2, there are at least ϵn strongly connected components, each containing fewer than x vertices. The probability that an arbitrarily chosen vertex belongs to one of these ϵn strongly connected components is at least $\frac{\epsilon n}{n} = \epsilon$. Our algorithm outputs ACCEPT for a NO instance only when the while-loop executes m times. The probability is at most $(1-\epsilon)^m = (1-\epsilon)^{\lceil \log_{(1-\epsilon)} \frac{1}{3} \rceil} \leq 1/3$. Hence, G is rejected with probability at least $2/3$.

Finding all of a vertex's incoming (outgoing) neighbors needs at most n queries on the incoming (outgoing, respectively) edges given the adjacency matrix. Since the while-loop is repeated at most m times, the query complexity is bounded by:

$$\begin{aligned}
& m \cdot 2 \cdot \left\lceil \frac{1}{n-2} \left(\frac{1}{\epsilon} - 1 \right) \right\rceil \cdot n < \\
& \left\lceil \log_{(1-\epsilon)} \frac{1}{3} + 1 \right\rceil \cdot \left\lceil \frac{2}{n-2} \left(\frac{1}{\epsilon} - 1 \right) + 2 \right\rceil \cdot n < \\
& \left\lceil \frac{1}{\log_{\frac{1}{3}} (1-\epsilon)} + 1 \right\rceil \cdot \left\lceil \frac{2}{n-2} \left(\frac{1}{\epsilon} - 1 \right) + 2 \right\rceil \cdot n = \\
& O \left(\frac{1}{\epsilon \log(1-\epsilon)} \right). \quad \blacksquare
\end{aligned}$$

Acknowledgments

The authors are grateful to Chin-Luei Chang and Yu-Chieh Chang for their suggestions.

References

- [1] Noga Alon and Asaf Shapira. “Testing Subgraphs in Directed Graphs.” *STOC*, 2003, pp. 700–709.
- [2] Noga Alon and Asaf Shapira. “A Characterization of Easily Testable Induced Subgraphs.” *SODA*, 2004, pp. 942–951.
- [3] B. Bollobas. “Complete Subgraphs are Elusive.” *J. Comb. Theory B*, 21, 1976, pp. 1-7.
- [4] Noga Alon, Eldar Fischer, Michael Krivelevich and Mario Szegedy. “Efficient Testing of Large Graphs.” *FOCS*, 1999, pp. 656–666.
- [5] Catherine E. Chronaki. “A Survey of Evasiveness: Lower Bounds on the Decision-Tree Complexity of Boolean Functions.” Available on <http://www.ics.forth.gr/chronaki/papers/ur/eve.ps> (2002).
- [6] Yevgeniy Dodis, Oded Goldreich, Eric Lehman, Sofya Raskhodnikova, Dana Ron and Alex Samorodnitsky. “Improved Testing Algorithms for Monotonicity.” *RANDOM-APPROX*, 1999, pp. 97–108.
- [7] Funda Ergun, Sampath Kannan, Ravi Kumar, Ronitt Rubinfeld and Mahesh Viswanathan. “Spot-Checkers.” *STOC*, 1998, pp. 259–268.
- [8] Eldar Fischer and Lance Fortnow. “Tolerant Versus Intolerant Testing for Boolean Properties.” *Electronic Colloquium on Computational Complexity (ECCC)*, 11(105), 2004.
- [9] Eldar Fischer and Ilan Newman. “Testing versus Estimation of Graph Properties.” *STOC*, 2005, pp. 138–146.
- [10] Zoltan Füredi, Assaf Naor and Jacques Verstraëte. “On the Turan Number for the Hexagon.” Available on <http://research.microsoft.com/research/theory/naor/homepage%20files/final-hexagons.pdf>, 2004.
- [11] Oded Goldreich and Dana Ron. “A Sublinear Bipartiteness Tester for Bounded Degree Graphs.” *Combinatorica*, 19(3), 1999, pp. 335–373.
- [12] Oded Goldreich, Shafi Goldwasser and Dana Ron. “Property Testing and Its Connection to Learning and Approximation.” *J. ACM*, 45(4), 1998, pp. 653–750.
- [13] Oded Goldreich, Shafi Goldwasser, Eric Lehman, Dana Ron and Alex Samorodnitsky. “Testing Monotonicity.” *Combinatorica*, 20(3), 2000, pp. 301–337.
- [14] Richard C. Holt, Edward M. Reingold. “On the Time Required To Detect Cycles and Connectivity in Graphs.” *Mathematical Systems Theory*, 6(2), 1972, pp. 103-106.
- [15] Michael J. Kearns and Dana Ron. “Testing Problems with Sublearning Sample Complexity.” *J. Comput. Syst. Sci.*, 61(3), 2000, pp. 428–456.
- [16] David G. Kirkpatrick. “Determining Graph Properties from Matrix Representations.” *STOC*, 1974, pp. 84-90.
- [17] Tamas Kövari, Vera T. Sós and Paul Turán. “On a Problem of K. Zarankiewicz.” *Colloquium Math.*, 3, 1954, pp. 50–57.
- [18] Ronitt Rubinfeld and Madhu Sudan. “Robust Characterizations of Polynomials with

Applications to Program Testing.” *SIAM J. Comput.*, 25(2), 1996, pp. 252–271.